

Tariffs tomorrow at 4pm. Hedge the book tonight?

A \$500M US equity book on 1 April 2025, the day before the reciprocal tariffs. An arbitrage-free S&P 500 volatility surface, rebuilt every day from real SPY option quotes, prices four hedges and marks them through the crash, the pause and expiry.

0.47

average fit error,
vol points

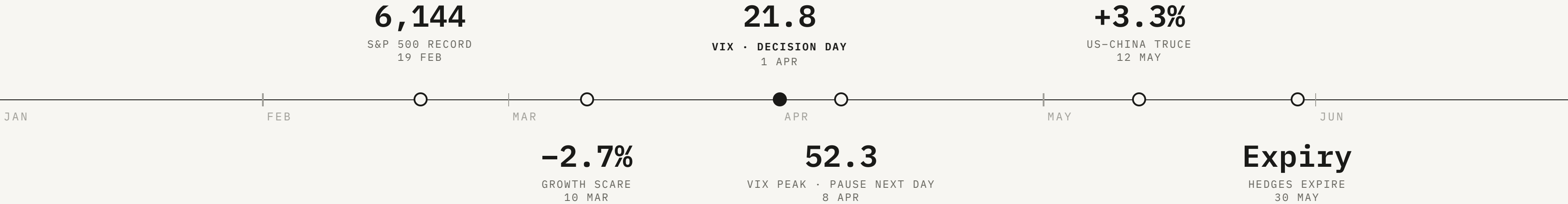
0.996

correlation with a VIX
the fit never sees

+\$33.2M

collar vs unhedged,
sold at VIX 45

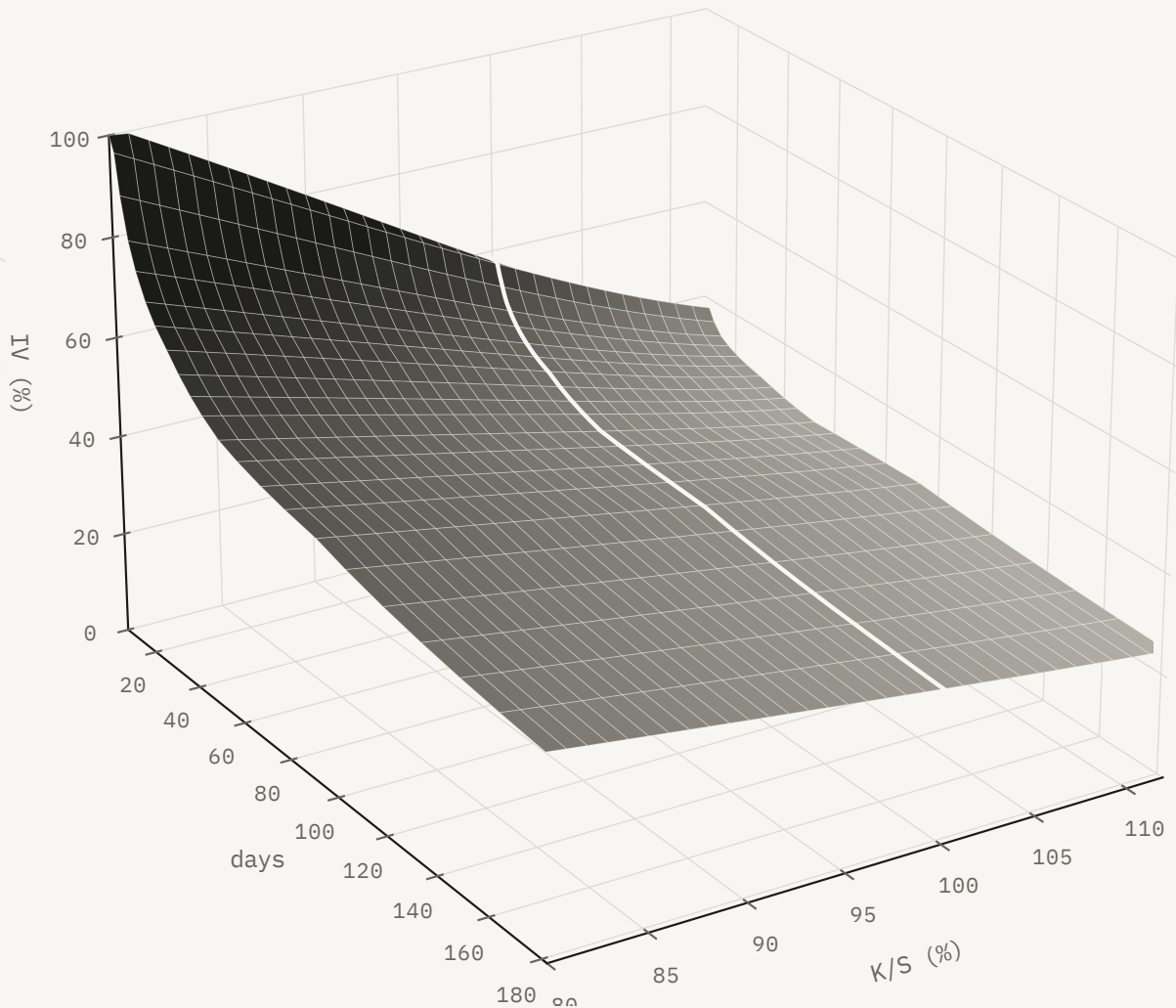
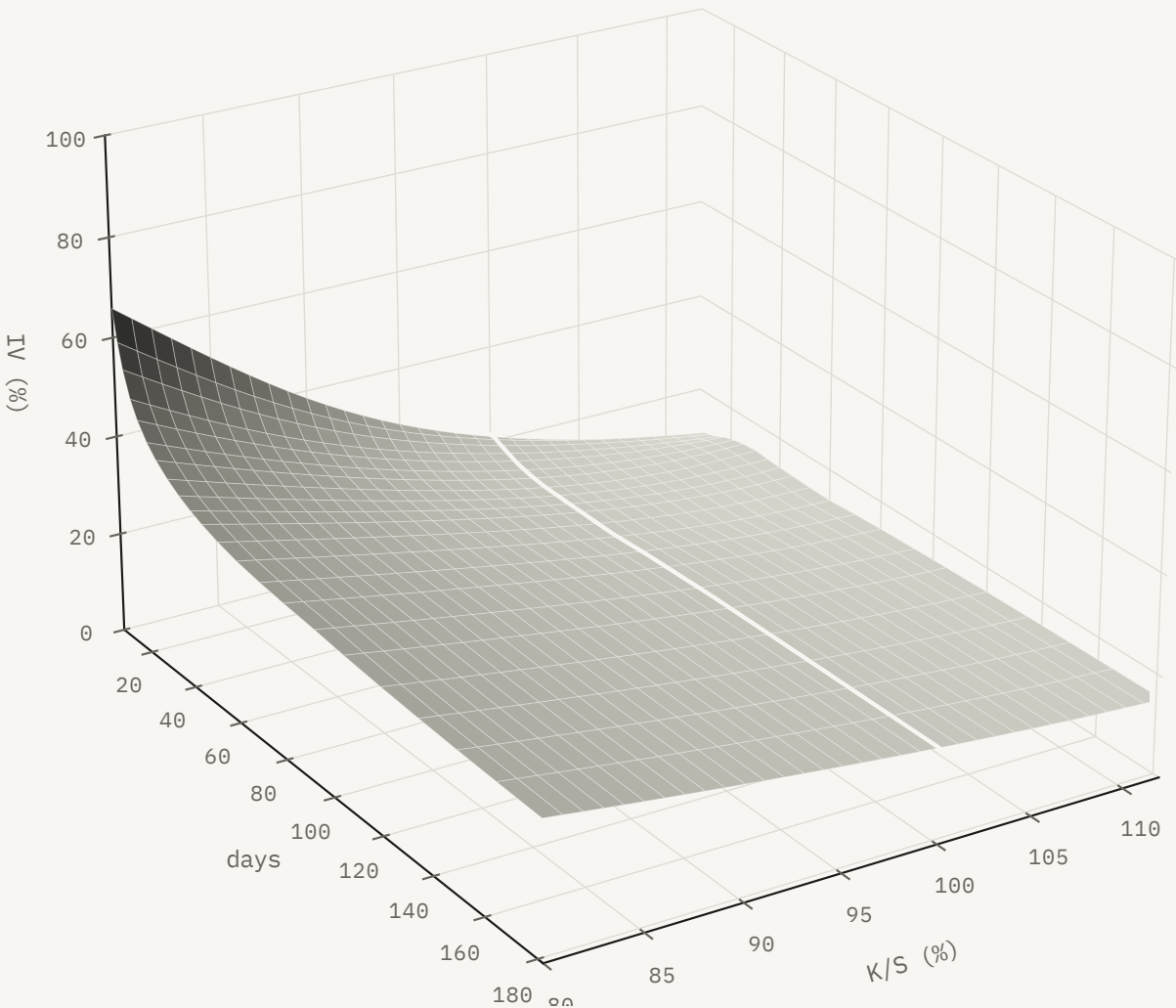
S&P 500 AND VIX, FIRST HALF OF 2025 · SOLID = DECISION DAY · HOLLOW = MARKET EVENTS



Four sessions later, short-dated vol had nearly tripled.

1 APRIL · S&P 500 5,633 · VIX 21.8

8 APRIL · S&P 500 4,983 · VIX 52.3



Implied vol by moneyness (strike / spot) and days to expiry, same scale on both days. The pale line traces the at-the-money term structure.

9-DAY VARIANCE · VIX9D

24.5 → 67.6

2.8× in four sessions

30-DAY ATM IMPLIED VOL

17.7% → 41.9%

S&P 500 down 12.1% from the 2 April close

25Δ RISK REVERSAL · 30D

-5.0 → -17.6

puts 52% vs calls 34% on 8 April

TERM STRUCTURE · VIX9D / VIX3M

1.12 → 1.63

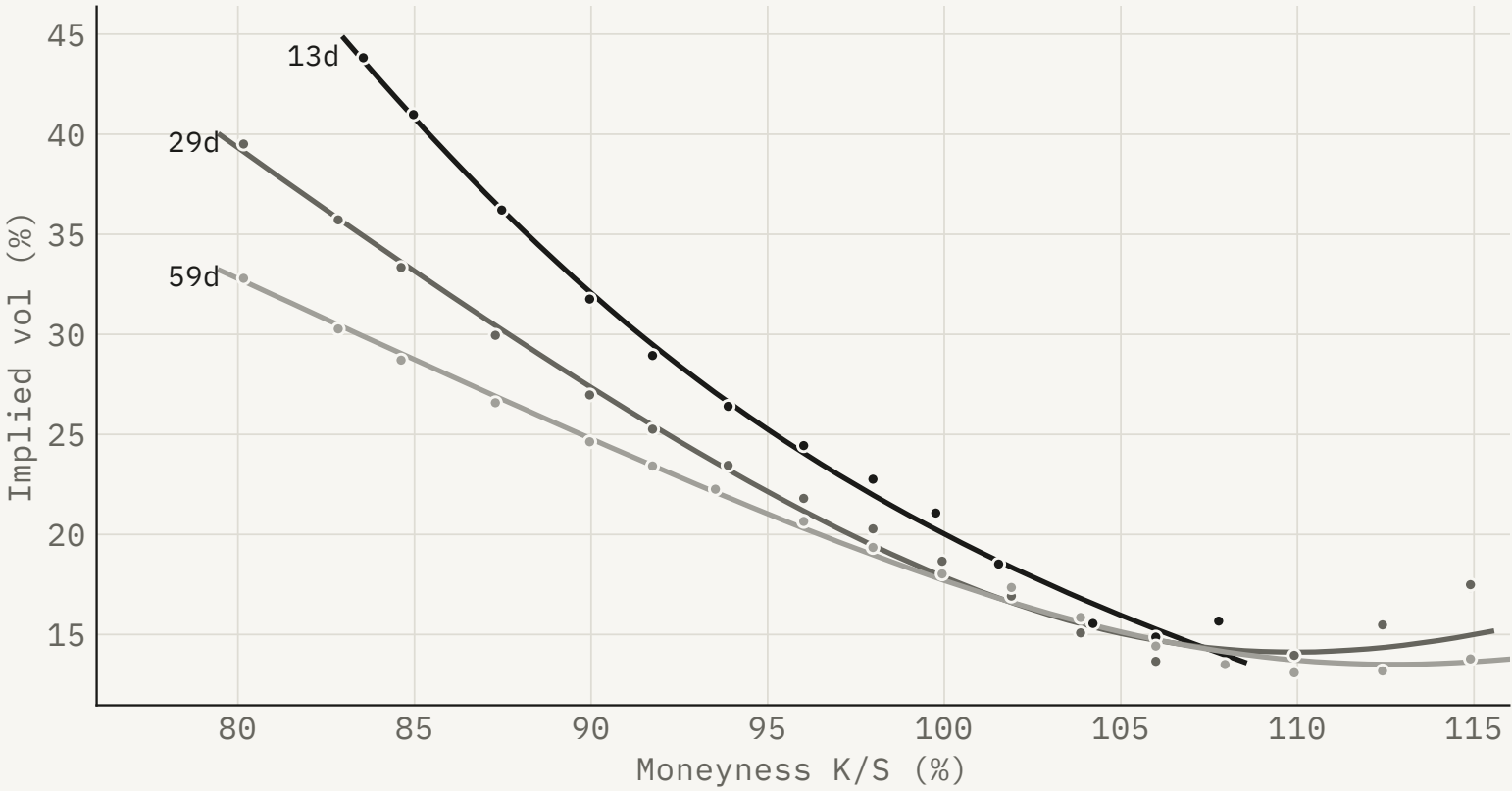
inverted on 1 April, deeply inverted on 8 April

Real bid/ask quotes in, an arbitrage-free surface out.

#	STEP	WHAT IT DOES	OUTPUT
1	Quotes	SPY end-of-day chains, bid and ask, 3 expiries a day	14,028 on 122 days
2	Spot and forwards	Put-call parity: quotes are not always taken at the close	28 Mar: 568 vs 555.66
3	Implied vols	Black-76 on out-of-the-money options, bid / mid / ask	6,166 quotes kept
4	SVI per expiry	Penalties: $g(k) \geq 0$, no crossing, Lee's wing bound	0.47 pts RMSE
5	Across maturities	Total-variance interpolation; $\sqrt{T}$ rule anchored to VIX9D, VIX3M, VIX6M	7 to 182 days

$g(k)$ is Gatheral's butterfly condition (a non-negative risk-neutral density). The 30-day VIX is left out of the fit on purpose.

1 APRIL · QUOTES WITH BID/ASK AND SVI FIT



TRADING DAYS

122

AVERAGE FIT ERROR

0.47 pts

BUTTERFLY VIOLATIONS

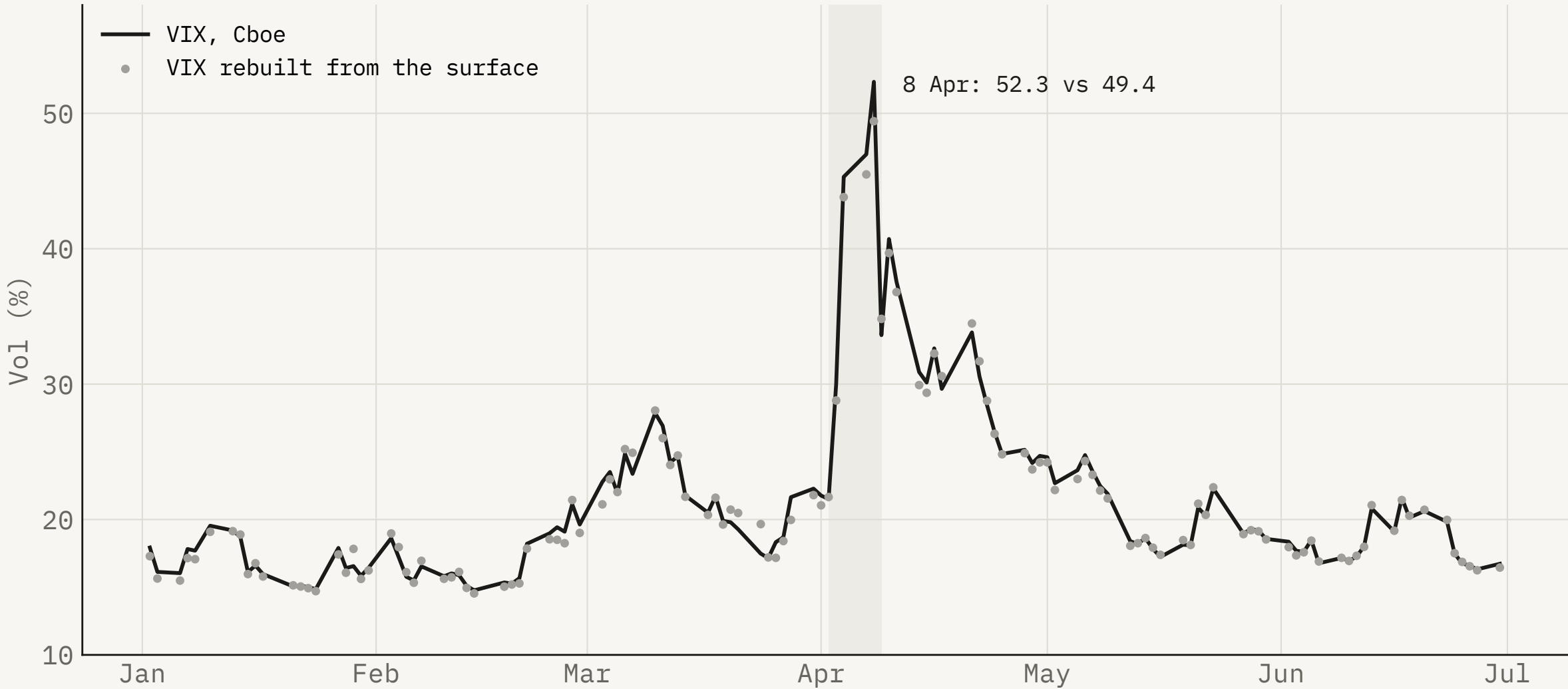
0 / 3.4M

CALENDAR BREACHES

0.015%

The surface rebuilds a VIX it never saw.

VIX, CBOE VS VIX REBUILT FROM THE SURFACE · 2 JANUARY TO 30 JUNE 2025



MEAN ERROR

-0.15 pts
median -0.12, standard deviation 0.63

CORRELATION WITH THE VIX

0.996
122 days, with the VIX from 14.8 to 52.3

WORST DAY

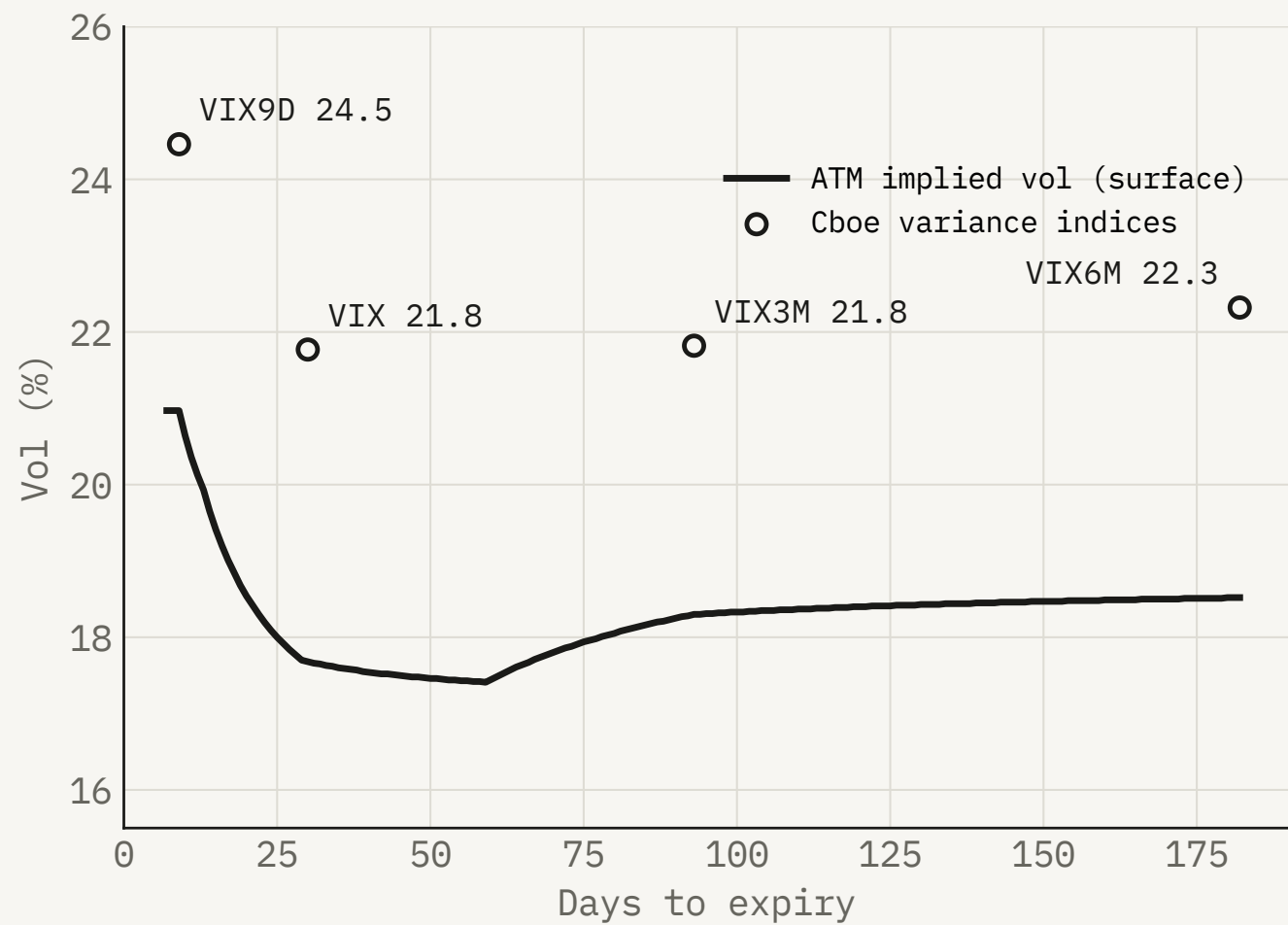
-2.9 pts
8 April: SPY vs SPX, tails beyond the quoted strikes

HOW IT IS REBUILT

$VS^2 \ T = 2 \int e^{-k_0 T M(k)} dk$
the continuous-strike variance swap behind the VIX

Protection was not expensive; the short end had priced the event.

TERM STRUCTURE, 1 APRIL



FOUR STRUCTURES ON THE 30 MAY EXPIRY · AT THE REAL QUOTES

STRUCTURE	LEGS	PREMIUM / SHARE	COST	% NAV
Long put 539	+539P	\$8.36	\$7.45M	1.49%
Put spread 539/475	+539P -475P	\$6.59	\$5.87M	1.17%
Collar 539/583	+539P -583C	\$1.40	\$1.25M	0.25%
Put-spread collar 539/475/583	+539P -475P -583C	-\$0.37	-\$0.33M	-0.07%

Buy at the ask, sell at the bid. 8,907 contracts cover the \$500M book at SPY 561.36. A negative premium is a credit.

30D ATM VS REALISED

17.7% vs 20.1%

25Δ RR VS H1 MEDIAN

-5.0 vs -4.5

VIX9D VS VIX

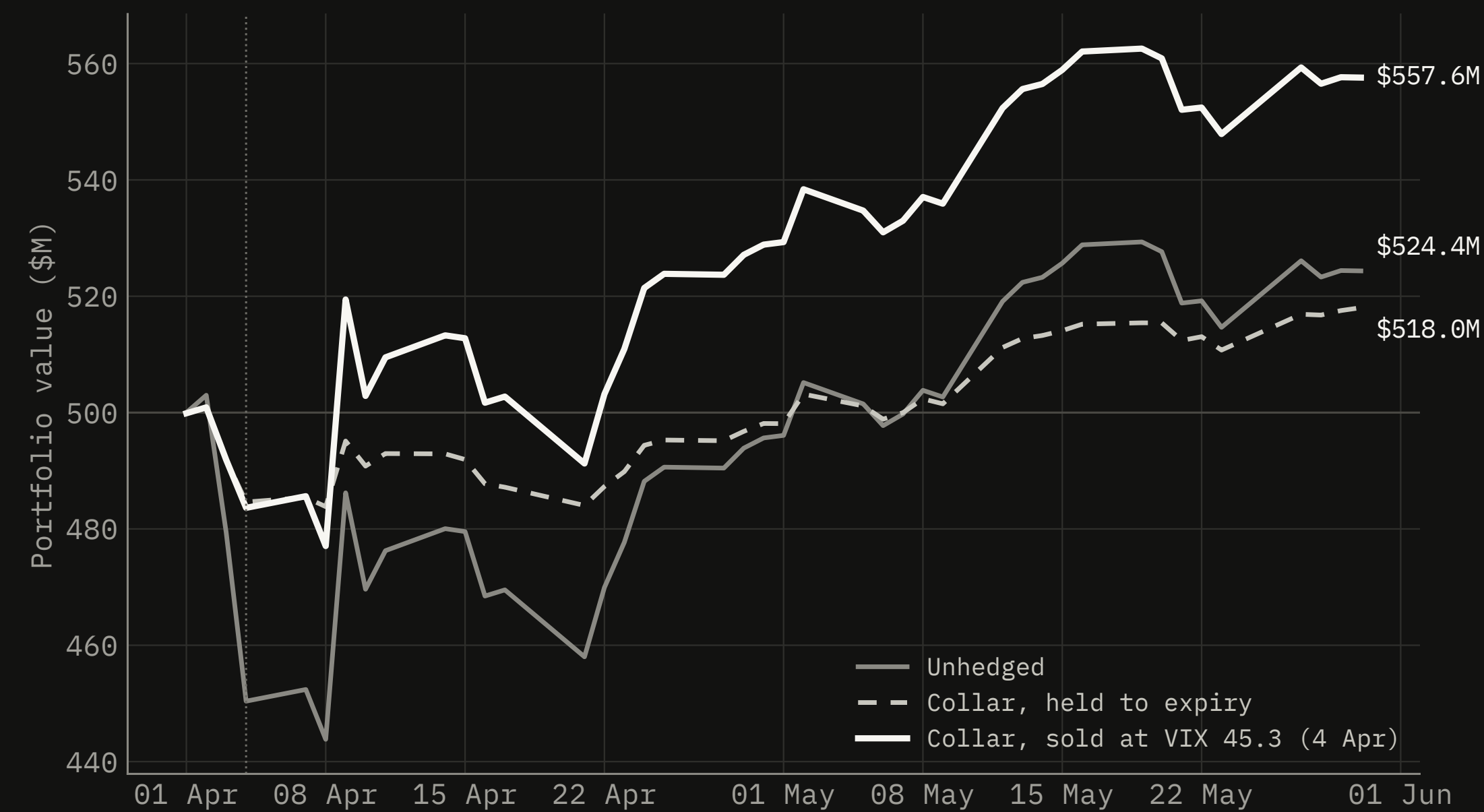
24.5 vs 21.8

CBOE SKEW

143

Held, the collar lost \$6M. Sold at VIX 45, it made \$33M.

PORTFOLIO VALUE, 1 APRIL TO 30 MAY · \$M



MAX DRAWDOWN · COLLAR HELD

-3.4%

vs -11.8% unhedged

COST ON 1 APRIL

0.25%

of NAV, \$1.25M for the collar

SOLD ON 4 APRIL FOR

\$34.5M

first close with the VIX above 40 (45.3)

VS UNHEDGED ON 30 MAY

+\$33.2M

-\$6.3M if held to expiry

Buy the collar, and decide in advance when to sell it.

FOUR STRUCTURES, TWO MANAGEMENT RULES · NAV IN \$M

STRUCTURE	COST, % NAV	MAX DD, HELD	30 MAY, HELD	30 MAY, SOLD	SOLD VS UNHEDGED
Unhedged	–	–11.8%	\$524.4M	\$524.4M	–
Long put 539	1.49%	–4.6%	\$516.9M	\$553.0M	+\$28.6M
Put spread 539/475	1.17%	–7.7%	\$518.5M	\$541.1M	+\$16.8M
Collar 539/583	0.25%	–3.4%	\$518.0M	\$557.6M	+\$33.2M
Put-spread collar 539/475/583	–0.07%	–6.5%	\$519.6M	\$545.7M	+\$21.4M

"Sold" = all legs closed at the first close with VIX ≥ 40 (4 April), at model mid minus half the quoted spread. Hedges marked daily on that day's fitted surface.

CONCLUSION

The collar gave the most protection per dollar: 0.25% of NAV for a drawdown of –3.4% instead of –11.8%. Held to expiry it still lost money once the pause and the truce lifted SPY above 583. The rule for selling the hedge decided the result, so it belongs in the trade ticket on day one.

Quotes fitted, 122 days6,166

VIX out of sample, mean error–0.15 pts

Marks vs quotes, 12 checks\$0.13

Deliverablesweb page · notebook · Excel · memo · deck